Worksheet 6, 15.3 and 6

1
2*integral(0-pi)(integral(4.3-7.8)(r*sqrt(7.8^2-r^2)drd(theta)))
= 577.216

Practice Problems

1. Janelle is creating beads. She starts with spheres of radius A and then drills out a hole in the shape of a cylinder of radius B where $B < A$. See `https://www.geogebra.org/3d/yqhhm3dv` What is the volume of the resulting bead?

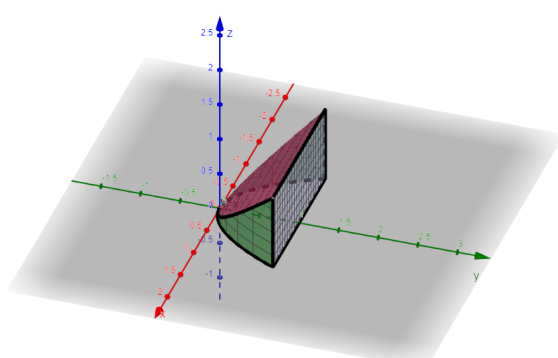
2. The solid below is bounded by $z = y$, $y = x^2$ and $y = 1$. Set up the integral to calculate its volume 4 different ways as indicated below. You just need to set up the integrals, not evaluate them. Interactive diagram: `https://www.geogebra.org/3d/hdsvuq2b`

2 $\int_0^1 \int_0^z \int_{\text{sqrt(y)}}^1 1\,dx\,dy\,dz$

2 $\int_0^1 \int_0^y \int_{\text{sqrt(y)}}^1 1\,dx\,dz\,dy$

2 $\int_0^1 \int_{\text{sqrt(y)}}^1 \int_0^y 1\,dz\,dx\,dy$

2 $\int_0^1 \int_{\text{x^2}}^1 \int_0^y 1\,dz\,dy\,dx$

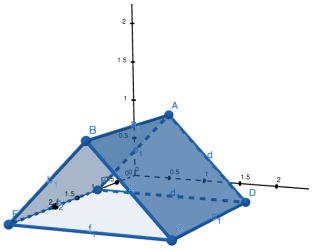


3. Evaluate
$$\iint_D e^{4-x^2-y^2}\,dA$$
where D is the region bounded by the semicircle $x = -\sqrt{4-y^2}$ and the $y-$axis.

3
convert to polar
2pi*integral(0-4)(re^(4-r^2))drd(theta)
= -pi(1-e^4)

Extra Quiz/Exam Practice

4. Use polar coordinates to find the volume of the solid bounded by the paraboloids $z = 3x^2 + 3y^2$ and $z = 4 - x^2 - y^2$.



5. The figure above shows the region in the first octant between $x = 1$, $x = 3$, $y = z$ and $y + z = 2$. The volume of this region can be calculated in two of the three ways below using a single iterated triple integral. Fill out the bounds on those two iterated triple integrals. Explain why you can't use a single triple integral for the third. Do NOT evaluate either iterated integral.

$$\int \quad \int \quad \int \quad 1\,dx\,dy\,dz$$

$$\int \quad \int \quad \int \quad 1\,dy\,dx\,dz$$

$$\int \quad \int \quad \int \quad 1\,dz\,dy\,dx$$

6. Evaluate $\iiint_E y\,dV$, where E is bounded by the planes $x = 0, y = 0, z = 0$ and $2x + 3y + 6z = 12$.

7. Set up a double integral to find the volume inside the cone $z = \sqrt{x^2 + y^2}$ and below the sphere
$$x^2 + y^2 + z^2 = R^2.$$

Challenge Questions

8. Revisit the bead question in number 1. Show that the volume of the resulting bead only depends on its height, not on A and B.

9. Pick a region of your choice. Find the area of the region using a double integral with both rectangular and polar coordinates. (You must use both coordinates on the same region - choose wisely!)

10. Pick your favorite integral from number 2 and evaluate it to find the volume of the pictured region.

11. Set up the triple integral in number 2 in the other two orders. (One requires 2 integrals and the other requires 3)

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